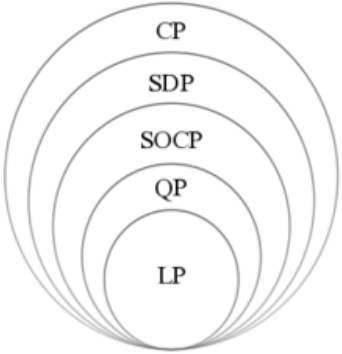
Optimization in Machine Learning

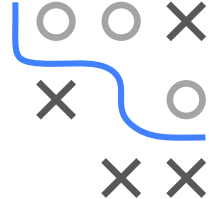
Constrained Optimization
Linear Programming



Learning goals

- Definition and different forms of an LP
- Geometric intuition of LPs
- Characteristics of vertices
- Simplex algorithm

LINEAR PROGRAMMING



- **Linear program** (LP): optimization problem with **linear** objective function + **linear** constraints
- General form

$$\min_{\boldsymbol{x}} \mathbf{c}^T \boldsymbol{x} \quad \text{s.t. } \mathbf{A}_1 \boldsymbol{x} \leq \mathbf{b}_1, \ \mathbf{A}_2 \boldsymbol{x} \geq \mathbf{b}_2, \ \mathbf{A}_3 \boldsymbol{x} = \mathbf{b}_3$$

- Examples

$$\min_{\boldsymbol{x}} \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|_1 \Leftrightarrow \quad \min_{\boldsymbol{x},\mathbf{s}} \mathbf{1}^T \mathbf{s} \quad \text{s.t. } \pm(\mathbf{A}\boldsymbol{x} - \mathbf{b}) \leq \mathbf{s}$$

$$\min_{\boldsymbol{x}} \|\mathbf{A}\boldsymbol{x} - \mathbf{b}\|_\infty \Leftrightarrow \quad \min_{\boldsymbol{x},t} t \quad \text{s.t. } \pm(\mathbf{A}\boldsymbol{x} - \mathbf{b}) \leq t\mathbf{1}$$

LINEAR PROGRAMMING: STANDARD FORM

Standard Form

$$\min \mathbf{c}^T \boldsymbol{x} \quad \text{s.t. } \mathbf{A}\boldsymbol{x} \leq \mathbf{b}, \ \boldsymbol{x} \geq \mathbf{0}$$

- $\mathbf{A}\boldsymbol{x} \geq \mathbf{b} \Leftrightarrow -\mathbf{A}\boldsymbol{x} \leq -\mathbf{b}$
- $\mathbf{A}\boldsymbol{x} = \mathbf{b} \Leftrightarrow \mathbf{A}\boldsymbol{x} \leq \mathbf{b}, \ -\mathbf{A}\boldsymbol{x} \leq -\mathbf{b}$
- $\boldsymbol{x} = \boldsymbol{x}^+ - \boldsymbol{x}^-$, where $\boldsymbol{x}^+ \geq \mathbf{0}, \ \boldsymbol{x}^- \geq \mathbf{0}$
- $\min_{\boldsymbol{x}} \mathbf{c}^T \boldsymbol{x} \Leftrightarrow \min_{\boldsymbol{x}^+, \boldsymbol{x}^-} \begin{bmatrix} \mathbf{c}^T & -\mathbf{c}^T \end{bmatrix} \begin{bmatrix} \boldsymbol{x}^+ \\ \boldsymbol{x}^- \end{bmatrix}$

LINEAR PROGRAMMING: EQUALITY FORM

- Equality Form

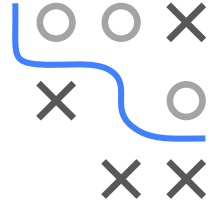
$$\min \mathbf{c}^T \boldsymbol{x} \quad \text{s.t. } \mathbf{A}\boldsymbol{x} = \mathbf{b},\ \boldsymbol{x} \geq \mathbf{0}$$

- By introducing slack variables $\mathbf{s}$:

$$\mathbf{A}\boldsymbol{x} \leq \mathbf{b} \Leftrightarrow \begin{bmatrix} \mathbf{A} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \mathbf{s} \end{bmatrix} = \mathbf{A}\boldsymbol{x} + \mathbf{s} = \mathbf{b},\ \mathbf{s} \geq 0$$

- $\min_{\boldsymbol{x}} \mathbf{c}^T \boldsymbol{x} \Leftrightarrow \min_{\boldsymbol{x},\mathbf{s}} \begin{bmatrix} \mathbf{c}^T & \mathbf{0}^T \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \mathbf{s} \end{bmatrix}$

GEOMETRIC INTERPRETATION



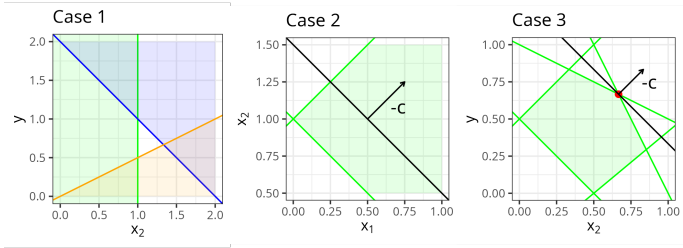
Feasible set

- Points $\{\boldsymbol{x} : \mathbf{A}_i^T \boldsymbol{x} = b_i\}$ form a hyperplane in $\mathbb{R}^n$
  $\mathbf{A}_i$ is perpendicular to the hyperplane and called **normal vector**
- Points $\{\boldsymbol{x} : \mathbf{A}_i^T \boldsymbol{x} \leq b_i\}$ lie on one side of the hyperplane, which form a convex half-space
- Points satisfying **all** inequalities form a **convex polytope**
  The intersection of convex sets is still convex

- Polytope $\{\boldsymbol{x} : \mathbf{A}\boldsymbol{x} \leq \mathbf{b},\ \mathbf{A} \in \mathbb{R}^{m \times n}\}$ is an n-**simplex**, i.e., convex hull of $n+1$ **affinely independent** points, which we call vertices

GEOMETRIC INTERPRETATION

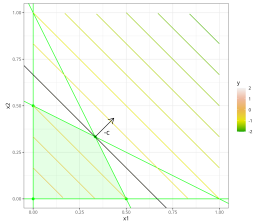
- There are 3 conditions for solving linear programming:
  1. Feasible set is **empty** $\Rightarrow$ LP is infeasible
  2. Feasible set is **unbounded**
  3. Feasible set is **bounded** $\Rightarrow$ LP has at least one solution



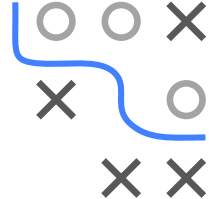
GEOMETRIC INTERPRETATION

Case 3: LP is feasible and bounded

- Points on the interior: never optimal, can be improved by moving along $-\mathbf{c}$
- Points on faces/edges: can be optimal only if the face/edge is perpendicular to $\mathbf{c}$
- Points on faces/edges not perpendicular to $\mathbf{c}$: can be improved by moving along $-\mathbf{c}$
- Vertices: can also be optimal

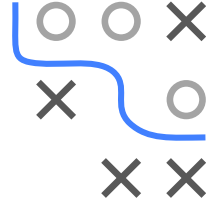


VERTICES



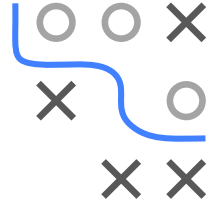
- Assume rows of $\mathbf{A} \in \mathbb{R}^{m \times n}$ are linearly independent and $m \leq n$ to form a bounded non-empty feasible set
- $\mathbf{A}\boldsymbol{x} = \mathbf{b}$ imposes m equality constraints:
  - Each equality constraint reduces the dimension of the feasible set by 1
  - Starting with n-dim space, applying m independent equality constraints leaves a solution space of dim $n - m$
- $\boldsymbol{x} \geq \mathbf{0}$ imposes n non-negativity constraints

VERTICES



- While satisfying $\mathbf{A}\boldsymbol{x} = \mathbf{b}$, the indices of a vertex vector can be partitioned into two sets:
  - $\mathcal{V}$ with $n - m$ elements: $i \in \mathcal{V} \Rightarrow x_i = 0$ (active constraints)
  - $\mathcal{B}$ with m elements: $i \in \mathcal{B} \Rightarrow x_i \geq 0$
- We have $\mathbf{A}_{\mathcal{B}}^{m \times m} \boldsymbol{x}_{\mathcal{B}} = \mathbf{b} \Rightarrow \boldsymbol{x}_{\mathcal{B}} = \mathbf{A}_{\mathcal{B}}^{-1} \mathbf{b}$
- **Note:** While every vertex has an associated partition $(\mathcal{B}, \mathcal{V})$, not every partition corresponds to a vertex

SIMPLEX ALGORITHM



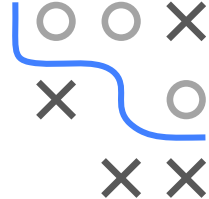
- The **simplex algorithm** solves LPs by moving from vertex to vertex of the feasible set, and produces an optimal vertex

- Operates on equality-form LPs $\mathbf{A}\boldsymbol{x} = \mathbf{b}, \boldsymbol{x} \geq \mathbf{0}$
  Assume rows of $\mathbf{A} \in \mathbb{R}^{m \times n}$ are linearly independent and $m \leq n$

- Guaranteed to arrive at an optimal solution so long as the LP is feasible and bounded

- The simplex algorithm operates in two phases:
  - **Initialization** phase: identifies a vertex partition
  - **Optimization** phase: transitions between vertex partitions toward a partition corresponding to an optimal vertex

SIMPLEX ALGORITHM



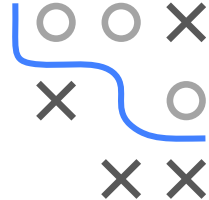
- Construct a Lagrangian for the equality form:

$$L(\boldsymbol{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) = \mathbf{c}^T \boldsymbol{x} - \boldsymbol{\mu}^T \boldsymbol{x} - \boldsymbol{\lambda}^T (\mathbf{A}\boldsymbol{x} - \mathbf{b})$$

  with $\boldsymbol{\mu} \geq \mathbf{0}$
- Optimal solution satisfies $\frac{\partial L}{\partial \boldsymbol{x}} = 0$, i.e., $\mathbf{A}^T \boldsymbol{\lambda} + \boldsymbol{\mu} = \mathbf{c}$
- Decompose stationarity condition into $\mathcal{B}$ and $\mathcal{V}$ components:

$$\mathbf{A}^T \boldsymbol{\lambda} + \boldsymbol{\mu} = \mathbf{c} \quad \Longrightarrow \quad \begin{cases} \mathbf{A}_{\mathcal{B}}^T \boldsymbol{\lambda} + \boldsymbol{\mu}_{\mathcal{B}} = \mathbf{c}_{\mathcal{B}} \\ \mathbf{A}_{\mathcal{V}}^T \boldsymbol{\lambda} + \boldsymbol{\mu}_{\mathcal{V}} = \mathbf{c}_{\mathcal{V}} \end{cases}$$

SIMPLEX ALGORITHM



- Choose $\boldsymbol{\mu}_{\mathcal{B}} = 0$ to satisfy $\boldsymbol{\mu} \odot \boldsymbol{x} = 0$, since for optimality, we need $\mu_i = 0$ when $x_i > 0$
- Compute $\boldsymbol{\lambda}$ from $\mathcal{B}$:

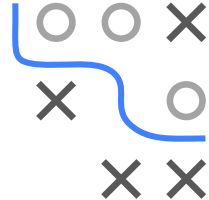
$$\mathbf{A}_{\mathcal{B}}^T \boldsymbol{\lambda} + \underbrace{\boldsymbol{\mu}_{\mathcal{B}}}_{=0} = \mathbf{c}_{\mathcal{B}} \quad \Longrightarrow \quad \boldsymbol{\lambda} = \mathbf{A}_{\mathcal{B}}^{-T} \mathbf{c}_{\mathcal{B}}$$

- Use this to obtain:

$$\begin{aligned} \mathbf{A}_{\mathcal{V}}^T \boldsymbol{\lambda} + \boldsymbol{\mu}_{\mathcal{V}} &= \mathbf{c}_{\mathcal{V}} \\ \boldsymbol{\mu}_{\mathcal{V}} &= \mathbf{c}_{\mathcal{V}} - \mathbf{A}_{\mathcal{V}}^T \boldsymbol{\lambda} \\ \boldsymbol{\mu}_{\mathcal{V}} &= \mathbf{c}_{\mathcal{V}} - \left(\mathbf{A}_{\mathcal{B}}^{-1} \mathbf{A}_{\mathcal{V}}\right)^T \mathbf{c}_{\mathcal{B}} \end{aligned}$$

- Knowing $\boldsymbol{\mu}_{\mathcal{V}}$ allows us to assess optimality of the vertices
  If $\boldsymbol{\mu}_{\mathcal{V}}$ contains negative components, then $\boldsymbol{\mu} \geq \mathbf{0}$ is not satisfied and the vertex is suboptimal

SIMPLEX ALGORITHM: OPTIMIZATION PHASE

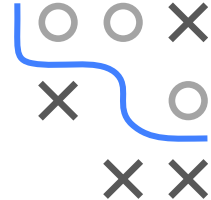


- Partition can be updated by swapping indices between $\mathcal{B}$ and $\mathcal{V}$
  Such a swap equates to moving from one vertex along an edge of the feasible set polytope to another vertex
- A transition $\boldsymbol{x} \to \boldsymbol{x}'$ between vertices must satisfy $\mathbf{A}\boldsymbol{x}' = \mathbf{b}$
- Starting with a partition defined by $\mathcal{B}$, choose an **entering index** $q \in \mathcal{V}$ that is to enter $\mathcal{B}$ using one of the heuristics described below
- The new vertex $\boldsymbol{x}'$ must satisfy:

$$\mathbf{A}\boldsymbol{x}' = \mathbf{A}_{\mathcal{B}}\boldsymbol{x}'_{\mathcal{B}} + \mathbf{A}_{\{q\}}x'_q = \mathbf{A}_{\mathcal{B}}\boldsymbol{x}_{\mathcal{B}} = \mathbf{A}\boldsymbol{x} = \mathbf{b}$$

- One **leaving index** $p \in \mathcal{B}$ in $\boldsymbol{x}_{\mathcal{B}}$ becomes zero during the transition, and is replaced by the column of $\mathbf{A}$ corresponding to index q
  This action is referred to as **pivoting**

SIMPLEX ALGORITHM: MINIMUM RATIO TEST



- Solve for the new design point:

$$\boldsymbol{x}'_{\mathcal{B}} = \boldsymbol{x}_{\mathcal{B}} - \mathbf{A}_{\mathcal{B}}^{-1}\mathbf{A}_{\{q\}}x'_q$$

- A particular **leaving index** $p \in \mathcal{B}$ becomes active when:

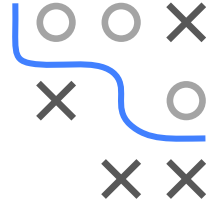
$$\left(\boldsymbol{x}'_{\mathcal{B}}\right)_p = 0 = \left(\boldsymbol{x}_{\mathcal{B}}\right)_p - \left(\mathbf{A}_{\mathcal{B}}^{-1}\mathbf{A}_{\{q\}}\right)_p x'_q$$

  and is thus obtained by increasing $x_q = 0$ to x'_q with:

$$x'_q = \frac{\left(\boldsymbol{x}_{\mathcal{B}}\right)_p}{\left(\mathbf{A}_{\mathcal{B}}^{-1}\mathbf{A}_{\{q\}}\right)_p}$$

- The leaving index is obtained using the **minimum ratio test**, which computes for each potential leaving index and selects the one with minimum x'_q
  We then swap p and q between $\mathcal{B}$ and $\mathcal{V}$

SIMPLEX ALGORITHM: EFFECT ON OBJECTIVE

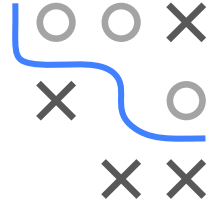


- Effect of a transition on the obj. function can be computed using x'_q
- Objective function value at the new vertex:

$$
\begin{aligned}
\mathbf{c}^T \boldsymbol{x}' &= \mathbf{c}_{\mathcal{B}}^T \boldsymbol{x}'_{\mathcal{B}} + c_q x'_q \\
&= \mathbf{c}_{\mathcal{B}}^T \left(\boldsymbol{x}_{\mathcal{B}} - \mathbf{A}_{\mathcal{B}}^{-1} \mathbf{A}_{\{q\}} x'_q \right) + c_q x'_q \\
&= \mathbf{c}_{\mathcal{B}}^T \boldsymbol{x}_{\mathcal{B}} - \mathbf{c}_{\mathcal{B}}^T \mathbf{A}_{\mathcal{B}}^{-1} \mathbf{A}_{\{q\}} x'_q + c_q x'_q \\
&= \mathbf{c}_{\mathcal{B}}^T \boldsymbol{x}_{\mathcal{B}} - (c_q - \mu_q) x'_q + c_q x'_q \\
&= \mathbf{c}^T \boldsymbol{x} + \mu_q x'_q
\end{aligned}
$$

using $\boldsymbol{\lambda} = \mathbf{A}_{\mathcal{B}}^{-T} \mathbf{c}_{\mathcal{B}}$ and $\mathbf{A}_{\{q\}}^T \boldsymbol{\lambda} = c_q - \mu_q$

SIMPLEX ALGORITHM: OPTIMALITY CONDITION

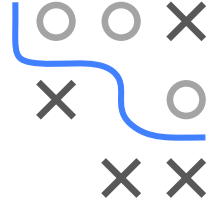


- Choosing an entering index q decreases the obj. function value by

$$\mathbf{c}^T \boldsymbol{x}' - \mathbf{c}^T \boldsymbol{x} = \mu_q x'_q$$

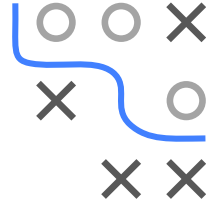
- The objective function decreases only if μ_q is negative
- To progress toward optimality, we must choose an index $q \in \mathcal{V}$ such that μ_q is negative
- If all components of $\boldsymbol{\mu}_{\mathcal{V}}$ are non-negative, we have found a global optimum

SIMPLEX ALGORITHM: HEURISTICS



- Since there can be multiple negative entries in $\boldsymbol{\mu}_{\mathcal{V}}$, different heuristics can be used to select an entering index:
- **Greedy heuristic**: choose a q that maximally reduces $\mathbf{c}^T\boldsymbol{x}$
- **Dantzig's rule**: choose the q with the most negative entry in $\boldsymbol{\mu}$
  Easy to calculate, but does not guarantee the maximum reduction in $\mathbf{c}^T\boldsymbol{x}$
  Also sensitive to scaling of the constraints
- **Bland's rule**: choose the first q with a negative entry in $\boldsymbol{\mu}$
  Tends to result in poor performance in practice when used alone
  Helps prevent cycles (returning to a visited vertex without decreasing objective)
  Usually used only after no improvements have been made for several iterations to break out of a cycle and ensure convergence

SIMPLEX ALGORITHM: EXAMPLE



$$\mathbf{A}_{\mathcal{V}} = \begin{pmatrix} 2 & 1 & 1 & 0 \\ -4 & 2 & 0 & 1 \end{pmatrix}, \mathbf{b} = (9,2)^T, \mathbf{c} = (3,-1,0,0)^T$$

Solution:

$\mathcal{V} = \{1,2\}, \mathcal{B} = \{3,4\}$

$\boldsymbol{x}_{\mathcal{B}} = \mathbf{A}_{\mathcal{B}}^{-1}\mathbf{b} = (9,2)^T$

$\boldsymbol{\lambda} = \mathbf{A}_{\mathcal{B}}^{-1}\mathbf{c}_{\mathcal{B}} = \mathbf{0}$

$\boldsymbol{\mu}_{\mathcal{V}} = \mathbf{c}_{\mathcal{V}} - \left(\mathbf{A}_{\mathcal{B}}^{-1}\mathbf{A}_{\mathcal{V}}\right)^T \mathbf{c}_{\mathcal{B}} = (3,-1)^T$

$\boldsymbol{\mu}_{\mathcal{V}}$ contains negative elements, so our current $\mathcal{B}$ is suboptimal.

We will pivot on the negative one with $q = 2, -\mathbf{A}_{\mathcal{B}}^{-1}\mathbf{A}_{\{q\}} = (1,2)^T$.
This causes $x_4 = 0$, so updated $\mathcal{B} = \{2,3\}$.
In the second iteration, we find $\boldsymbol{x}_{\mathcal{B}} = (1,8)^T, \boldsymbol{\lambda} = (0,-\frac{1}{2})^T, \boldsymbol{\mu}_{\mathcal{V}} = (1,\frac{1}{2})^T$.

This is optimal with no negative entry, thus we have $x^* = (0,1,8,0)^T$